

* $\longrightarrow$ sign refers
to a transformation

1.1

9. Given a system of the form

$$-m_1x_1 + x_2 = b_1$$

$$-m_2x_1 + x_2 = b_2$$

where m_1, m_2, b_1 , and b_2 are constants:

(a) Show that the system will have a unique solution if

$$m_1 \neq m_2.$$

(b) Show that if $m_1 = m_2$, then the system will be consistent only if $b_1 = b_2$.

(c) Give a geometric interpretation of parts (a) and (b).

(a)

$$\begin{array}{l} -m_1x_1 + x_2 = b_1 \\ -m_2x_1 + x_2 = b_2 \end{array} \longrightarrow \begin{bmatrix} -m_1 & 1 \\ -m_2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$\longrightarrow \left[\begin{array}{cc|c} -m_1 & 1 & b_1 \\ -m_2 & 1 & b_2 \end{array} \right] \xrightarrow{\text{Gaussian elimination}} \left[\begin{array}{cc|c} -m_1 & 1 & b_1 \\ 0 & 1 - \frac{m_2}{m_1} & b_2 - b_1 \cdot \frac{m_2}{m_1} \end{array} \right]$$

(a)

$$\text{For } m_1 \neq m_2, m_1 \neq 0, 1 - \frac{m_2}{m_1} \neq 0$$

$$x_2 \cdot \left(1 - \frac{m_2}{m_1}\right) = \left(b_2 - b_1 \frac{m_2}{m_1}\right)$$

$$\Rightarrow x_2 = \frac{\left(b_2 - b_1 \frac{m_2}{m_1}\right)}{\left(1 - \frac{m_2}{m_1}\right)}$$

$$\text{for } b_2 - b_1 \frac{m_2}{m_1} = 0$$

$$x_2 = 0, x_1 = \frac{b_1}{-m_1}$$

$$\text{for } b_2 - b_1 \frac{m_2}{m_1} \neq 0$$

$$x_2 \neq 0, x_1 = \frac{b_1 - x_2}{-m_1}$$

$\Rightarrow$ vector $\vec{x}$ has only solution *

(b) For $m_1 = m_2$, $m \neq 0$, $1 - \frac{m_2}{m_1} = 0$

when $b_1 = b_2$, $b_2 - b_1 \frac{m_2}{m_1} = 0$, and we get

$$\left[\begin{array}{cc|c} -m_1 & 1 & b_1 \\ 0 & 1 - \frac{m_2}{m_1} & b_2 - b_1 \frac{m_2}{m_1} \end{array} \right] \longrightarrow \left[\begin{array}{cc|c} -m_1 & 1 & b_1 \\ 0 & 0 & 0 \end{array} \right]$$

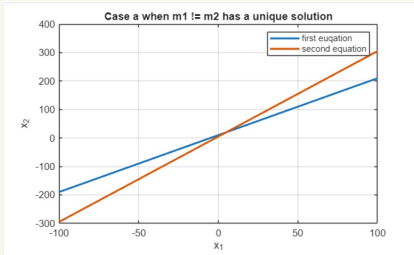
second row is always true, so x_2 is a free variable, set $x_2 = \alpha$

$x_1 = \frac{b_1 - \alpha}{-m_1} \Rightarrow$ we can see vector $\vec{x}$ has infinitely many solutions $\rightarrow$ consistent ~~✗~~

if $b_1 \neq b_2$, $b_2 - b_1 \frac{m_2}{m_1} \neq 0$, and second equation is always false
 $\Rightarrow$ inconsistent (no solutions)

(c)

For case (a)



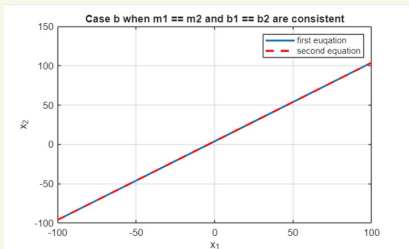
$(m_1 \neq 0)$

$m_1 \neq m_2$, set $m_1 = 1$, $m_2 = 3$

and set $b_1 = 5$, $b_2 = 10$

$\Rightarrow$ only one solution

For case (b)



$(m_1 \neq 0)$

$m_1 = m_2$, $b_1 = b_2$

set $m_1 = m_2 = 1$, $b_1 = b_2 = 4$

$\Rightarrow$ infinitely many solutions

6. Use Gauss-Jordan reduction to solve each of the following systems:

(a) $2x + y = 1$
 $7x + 6y = 1$

(b) $x_1 + x_2 - x_3 + x_4 = 6$
 $2x_1 - x_2 + x_3 - x_4 = -3$
 $3x_1 + x_2 - 2x_3 + x_4 = 9$

(c) $x_1 - 10x_2 + 5x_3 = -4$
 $x_1 + x_2 + x_3 = 1$

(d) $x_1 - 2x_2 + 3x_3 + x_4 = 4$
 $2x_1 + x_2 - x_3 + x_4 = 1$
 $x_1 + 3x_2 + x_3 + x_4 = 3$

(a)

$$\begin{cases} 2x + y = 1 \\ 7x + 6y = 1 \end{cases}$$

$$\rightarrow \left[\begin{array}{cc|c} 2 & 1 & 1 \\ 7 & 6 & 1 \end{array} \right] \xrightarrow{\text{row}_2 - \frac{7}{2}\text{row}_1} \left[\begin{array}{cc|c} 2 & 1 & 1 \\ 0 & \frac{5}{2} & -\frac{5}{2} \end{array} \right]$$

$$\xrightarrow{\text{row}_1 - \frac{2}{5}\text{row}_2} \left[\begin{array}{cc|c} 2 & 0 & 2 \\ 0 & \frac{5}{2} & -\frac{5}{2} \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -1 \end{array} \right]$$

$$\Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad x_1 = 1, \quad x_2 = -1$$

$$\begin{array}{l} \text{row}_1 \leftrightarrow \text{row}_2 \\ \text{row}_2 \leftrightarrow \text{row}_3 \end{array} \left[\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 6 \\ 2 & -1 & 1 & -1 & -3 \\ 3 & 1 & -2 & 1 & 9 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{-\frac{2}{3}\text{row}_1} \left[\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 6 \\ 0 & -3 & 3 & -3 & -15 \\ 0 & -2 & 1 & -2 & -9 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 6 \\ 0 & -3 & 3 & -3 & -15 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 6 \\ 0 & 1 & -1 & 1 & 5 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 7 \\ 0 & 1 & 0 & 1 & 6 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 4 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

row₄ is always zero $\Rightarrow x_4$ is free variable, $x_4 = \alpha$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ -1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 + x_4 \\ x_3 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ -1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ -1 \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 = 1$$

$$x_2 = 4 - \alpha$$

$$x_3 = -1$$

$$x_4 = \alpha$$

$$(c) \left[\begin{array}{ccc|c} 1 & -10 & 5 & -4 \\ 1 & 1 & 1 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & -10 & 5 & -4 \\ 0 & 11 & -4 & 5 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & -10 & 5 & -4 \\ 0 & 1 & -\frac{4}{11} & \frac{5}{11} \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 0 & \frac{15}{11} & \frac{6}{11} \\ 0 & 1 & -\frac{4}{11} & \frac{5}{11} \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & \frac{15}{11} & \frac{6}{11} \\ 0 & 1 & -\frac{4}{11} & \frac{5}{11} \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$x_3 = \alpha, \quad \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{6}{11} \\ \frac{5}{11} \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} -\frac{15}{11} \\ \frac{4}{11} \\ 0 \end{bmatrix}, \quad \begin{aligned} x_1 &= \frac{6}{11} - \frac{15}{11}\alpha \\ x_2 &= \frac{5}{11} + \frac{4}{11}\alpha \end{aligned}$$

$$x_3 = \alpha$$

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$$(d) \left[\begin{array}{cccc|c} 1 & -2 & 3 & 1 & 4 \\ 2 & 1 & -1 & 1 & 1 \\ 1 & 3 & 1 & 1 & 3 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & -2 & 3 & 1 & 4 \\ 0 & 5 & -7 & -1 & -7 \\ 0 & 5 & -2 & 0 & -1 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & -2 & 3 & 1 & 4 \\ 0 & 5 & -7 & -1 & -7 \\ 0 & 0 & 5 & 1 & 6 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & -2 & 3 & 1 & 4 \\ 0 & 1 & -\frac{7}{5} & -\frac{1}{5} & -\frac{7}{5} \\ 0 & 0 & 1 & \frac{1}{5} & \frac{6}{5} \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{\frac{2}{5}} \left[\begin{array}{cccc|c} 1 & -2 & 0 & \frac{2}{5} & \frac{2}{5} \\ 0 & 1 & 0 & \frac{2}{5} & \frac{7}{25} \\ 0 & 0 & 1 & \frac{1}{5} & \frac{6}{5} \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & \frac{14}{25} & \frac{24}{25} \\ 0 & 1 & 0 & \frac{2}{25} & \frac{7}{25} \\ 0 & 0 & 1 & \frac{1}{5} & \frac{6}{5} \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$x_4 = \alpha, \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{24}{25} \\ \frac{7}{25} \\ \frac{6}{5} \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} -\frac{14}{25} \\ -\frac{2}{25} \\ -\frac{1}{5} \\ 0 \end{bmatrix}$$

$$x_1 = \frac{24}{25} - \frac{14}{25}\alpha$$

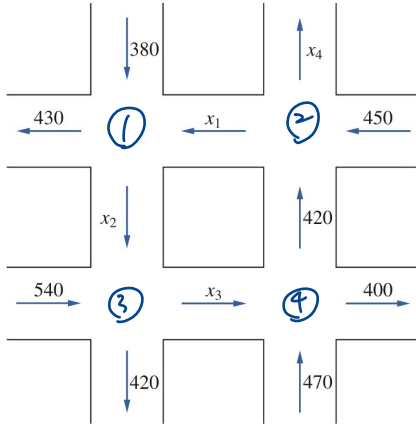
$$x_2 = \frac{7}{25} - \frac{2}{25}\alpha$$

$$x_3 = \frac{6}{5} - \frac{1}{5}\alpha$$

$$x_4 = \alpha$$

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15. Determine the values of x_1, x_2, x_3, x_4 for the following traffic flow diagram:



$$\begin{cases} x_1 + 380 = 430 + x_2 \\ 420 + 450 = x_1 + x_4 \\ x_2 + 540 = x_3 + 420 \\ x_3 + 470 = 400 + 420 \end{cases}$$

$$\Rightarrow \begin{cases} x_1 - x_2 = 50 \\ x_1 + x_4 = 870 \\ x_2 - x_3 = -120 \\ x_3 = 350 \end{cases}$$

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 50 \\ 1 & 0 & 0 & 1 & 870 \\ 0 & 1 & -1 & 0 & -120 \\ 0 & 0 & 1 & 0 & 350 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 50 \\ 0 & 1 & -1 & 0 & -120 \\ 0 & 0 & 1 & 0 & 350 \\ 1 & 0 & 0 & 1 & 870 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 50 \\ 0 & 1 & -1 & 0 & -120 \\ 0 & 0 & 1 & 0 & 350 \\ 0 & 1 & 0 & 1 & 820 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 50 \\ 0 & 1 & -1 & 0 & -120 \\ 0 & 0 & 1 & 0 & 350 \\ 0 & 0 & 1 & 1 & 940 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 50 \\ 0 & 1 & -1 & 0 & -120 \\ 0 & 0 & 1 & 0 & 350 \\ 0 & 0 & 0 & 1 & 590 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 50 \\ 0 & 1 & 0 & 0 & 230 \\ 0 & 0 & 1 & 0 & 350 \\ 0 & 0 & 0 & 1 & 590 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 280 \\ 0 & 1 & 0 & 0 & 230 \\ 0 & 0 & 1 & 0 & 350 \\ 0 & 0 & 0 & 1 & 590 \end{array} \right]$$

$$x_1 = 280$$

$$x_2 = 230$$

$$x_3 = 350$$

$$x_4 = 590$$

13. Let $Ax = b$ be a linear system whose augmented matrix $(A|b)$ has reduced row echelon form

$$\left(\begin{array}{ccccc|c} 1 & 2 & 0 & 3 & 1 & -2 \\ 0 & 0 & 1 & 2 & 4 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

(a) Find all solutions to the system.

(b) If

$$a_1 = \begin{pmatrix} 1 \\ 1 \\ 3 \\ 4 \end{pmatrix} \quad \text{and} \quad a_3 = \begin{pmatrix} 2 \\ -1 \\ 1 \\ 3 \end{pmatrix}$$

determine b .

(a) $x_2 = \alpha$

$x_4 = \beta$

$x_5 = \gamma$

$$\left[\begin{array}{ccccc} 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 1 & 2 & 4 \end{array} \right]$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -2 \\ \alpha \\ 5 \\ \beta \\ \gamma \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 + 2\alpha + 3\beta + \gamma \\ x_3 + \beta + 4\gamma \end{bmatrix} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 5 \end{bmatrix} + \alpha \begin{bmatrix} -2 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -3 \\ -1 \end{bmatrix} + \gamma \begin{bmatrix} -1 \\ -4 \end{bmatrix} \quad \text{✗}$$

(b)

$$A = \left[\begin{array}{c|c|c|c|c} | & | & | & | & | \\ a_1 & a_2 & a_3 & a_4 & a_5 \\ | & | & | & | & | \end{array} \right] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} | \\ b \\ | \end{bmatrix}$$

$$A = x_1 \begin{bmatrix} | \\ a_1 \\ | \end{bmatrix} + x_2 \begin{bmatrix} | \\ a_2 \\ | \end{bmatrix} + x_3 \begin{bmatrix} | \\ a_3 \\ | \end{bmatrix} + x_4 \begin{bmatrix} | \\ a_4 \\ | \end{bmatrix} + x_5 \begin{bmatrix} | \\ a_5 \\ | \end{bmatrix} = \begin{bmatrix} | \\ b \\ | \end{bmatrix}$$

when $x_2 = \alpha = 0$
 $x_4 = \beta = 0$
 $x_5 = \gamma = 0$

$$\Rightarrow \begin{matrix} x_1 = -2 \\ x_3 = 5 \end{matrix} \Rightarrow -2 \begin{bmatrix} | \\ a_1 \\ | \end{bmatrix} + 5 \begin{bmatrix} | \\ a_3 \\ | \end{bmatrix} = -2 \times \begin{bmatrix} | \\ 1 \\ 4 \end{bmatrix} + 5 \begin{bmatrix} | \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} | \\ b \\ | \end{bmatrix}$$

$$\Rightarrow b = \begin{bmatrix} 8 \\ -7 \\ -1 \\ 7 \end{bmatrix} \quad \text{✗}$$

程式作業

1. Use MATLAB to generate random 5×5 matrices A and B . For each of the following, compute A_1, A_2, A_3 , and A_4 as indicated, and determine which of the matrices are equal (you can use MATLAB to test whether two matrices are equal by computing their difference).

- (a) $A_1 = A * B, A_2 = B * A, A_3 = (A' * B)', A_4 = (B' * A)'$
 (b) $A_1 = A * B', A_2 = A' * B, A_3 = (B * A)', A_4 = (B' * A)'$
 (c) $A_1 = \text{inv}(A * B), A_2 = \text{inv}(A) * \text{inv}(B), A_3 = \text{inv}(B * A), A_4 = \text{inv}(B) * \text{inv}(A)$
 (d) $A_1 = \text{inv}(A * B'), A_2 = \text{inv}(A) * \text{inv}(B)', A_3 = \text{inv}(B) * \text{inv}(A), A_4 = (\text{inv}(A' * B))'$

question (a)

$A_1 = A_4$ are equal

$A_2 = A_3$ are equal

question (b)

$A_1 = A_3$ are equal

$A_2 = A_4$ are equal

question (c)

$A_1 = A_4$ are equal

$A_2 = A_3$ are equal

question (d)

$A_1 = A_3$ are equal

$A_2 = A_4$ are equal

手寫證明

(a) $A_1 = AB, A_2 = BA$
 $A_3 = (A'B')' = (B')'(A')' = BA$
 $A_4 = (B'A')' = AB$
 $\Rightarrow A_1 = A_4, A_2 = A_3$

(b) $A_1 = AB'$
 $A_2 = A'B$
 $A_3 = (BA')' = AB'$
 $A_4 = (B'A)' = A'B$

$\Rightarrow A_1 = A_3, A_2 = A_4$

(c) $A_1 = (AB)' = B'A'$
 $A_2 = A'B'$
 $A_3 = (BA)' = A'B'$
 $A_4 = B'A'$

$\Rightarrow A_1 = A_4, A_2 = A_3$

(d) $A_1 = (AB')' = BA'$
 $A_2 = A'(B')' = A'B$
 $A_3 = (B')A' = BA'$
 $A_4 = ((A'B)')' = A'B$

$\Rightarrow A_1 = A_3, A_2 = A_4$

